

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

3rd Semester of B. Tech. Examination (EC)

November 2013

## EE220 ELECTRICAL MACHINES AND POWER CIRCUITS

Date: 30.11.2013, Saturday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

### Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

### SECTION – I

- Q - 1 (a) Explain working principle of dc generator with necessary diagram. [07]
- Q - 2 (a) Enumerate methods of speed control of dc motor. Explain Ward Leonard method in brief. [07]
- (b) Explain torque speed characteristic of dc shunt motor. [07]

### OR

- Q - 2 (a) Define transformer and explain shell type and core type transformer. [07]
- (b) Explain conditions for parallel operation of transformer. [07]
- Q - 3 (a) Explain auto transformer in detail. [07]
- (b) Give the comparison of indoor and outdoor substation. [07]

### OR

- Q - 3 (a) Explain operation of transformer on load. [07]
- (b) Explain underground substation with necessary diagram. [07]

### SECTION – II

- Q - 4 (a) State the various methods for measurement of slip of induction motor and explain any one of them. [07]
- Q - 5 (a) Explain any one method for speed control of slip ring induction motor. [07]
- (b) Explain in brief the effect of change in supply voltage on performance of induction motor. [07]

### OR

- Q - 5 (a) Explain double field revolving theory of single phase induction motor. [07]
- (b) Explain shaded pole induction motor with necessary diagrams. [07]
- Q - 6 (a) Explain construction of alternator in brief. [07]
- (b) Explain synchronous impedance method for determination of regulation of alternator. [07]

### OR

Q - 6 (a) Explain types of tariffs in detail.

[07]

(b) Define power factor and explain disadvantages low power factor.

[07]

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